수열과 급수

연습문제 풀이

09

연습문제 9.2

1. 다음 급수의 수렴성을 조사하여라.

(3) $\sum_{n=0}^{\infty} \frac{n^2}{n+1}$

$\lim_{n\to\infty} \frac{n^2}{n+1} = \infty \quad \neq 0$

$\therefore$ 발산

(5) $\sum_{n=0}^{\infty} \sin\left(\frac{\pi}{2} - \frac{1}{n}\right)$

$\lim_{n\to\infty} \sin\left(\frac{\pi}{2} - \frac{1}{n}\right) = 1 \quad \neq 0$

$\therefore$ 발산

(10) $\sum_{n=1}^{\infty} (-1)^n (0.3)^n$

$\sum_{n=1}^{\infty} (-1)^n (0.3)^n = \sum_{n=1}^{\infty} (-0.3)^n$

정리9.6에의해 $|r = (-0.3)| < 1$ 이므로 수렴

연습문제 9.3

1. 다음 급수의 수렴성을 조사하여라.

(1) $\sum_{n=1}^{\infty}\frac{1}{(n+1)^2}$

정리 9.9에 의해

$$\int_1^{\infty}\frac{1}{(x+1)^2}dx=\lim_{m\to\infty}\int_1^m(x+1)^{-2}dx=\lim_{m\to\infty}\left[-\frac{1}{x+1}\right]_1^m$$

$$\lim_{m\to\infty}\left(-\frac{1}{m+1}+\frac{1}{2}\right)=\frac{1}{2}$$

$\therefore$ 수렴

(4) $\sum_{n=1}^{\infty}\frac{1}{n+\sqrt{n}}$

정리 9.10에 의해

$$\frac{1}{n+\sqrt{n}}\geq\frac{1}{n+n}$$

$$\sum_{n=1}^{\infty}\frac{1}{n+n}=\left(\frac{1}{2}\right)+\left(\frac{1}{4}\right)+\left(\frac{1}{6}\right)+\cdots+\left(\frac{1}{n+n}\right)=\frac{1}{2}\sum_{1}^{\infty}\frac{1}{n}$$

$\therefore$ 발산

연습문제 9.3

(11) $\sum_{n=1}^{\infty} \frac{\sqrt{n}}{n^2+3}$

정리 9.10에 의해

$$\frac{\sqrt{n}}{n^2+3} < \frac{\sqrt{n}}{n^2}$$

$$\sum_{n=1}^{\infty} \frac{\sqrt{n}}{n^2} = \sum_{n=1}^{\infty} \frac{1}{n^{\frac{3}{2}}}$$

$\frac{3}{2} \geq 1$ 이므로 수렴 (예제 9.12 참조)

연습문제 9.4

1. 다음 급수의 수렴성을 조사하여라.

(1) $\sum_{n=0}^{\infty} \frac{n!}{2^n}$

$$r = \lim_{n\to\infty} \frac{a_{n+1}}{a_n} = \lim_{n\to\infty} \frac{\frac{(n+1)!}{n^{n+1}}}{\frac{n!}{2^n}}$$

$$= \lim_{n\to\infty} \frac{2^n \cdot (n+1)!}{2^{n+1} \cdot n!}$$

$$= \lim_{n\to\infty} \frac{(n+1)}{2} = \infty$$

$\therefore$ 발산

(3) $\sum_{n=0}^{\infty} \frac{n!3^n}{10^n}$

$$r = \lim_{n\to\infty} \frac{a_{n+1}}{a_n} = \lim_{n\to\infty} \frac{\frac{(n+1)! \cdot 3^{n+1}}{10^{n+1}}}{\frac{n! \cdot 3^n}{10^n}}$$

$$= \lim_{n\to\infty} \frac{10^n \cdot (n+1)! \cdot 3^{n+1}}{10^{n+1} \cdot n! \cdot 3^n} = \lim_{n\to\infty} \frac{3(n+1)}{10} = \infty$$

$\therefore$ 발산

연습문제 9.4

(5) $\sum_{n=1}^{\infty}(\frac{n}{2n+5})^n$

$$\rho = \lim_{n\to\infty}(a_n)^{1/n} = \lim_{n\to\infty}\left\{(\frac{n}{2n+5})^n\right\}^{\frac{1}{n}}$$

$$= \lim_{n\to\infty}\frac{n}{2n+5} = \frac{1}{2} < 1$$

$\therefore$ 수렴

(10) $\sum_{n=1}^{\infty}\frac{n!}{e^n}$

$$r = \lim_{n\to\infty}\frac{\frac{(n+1)!}{e^{n+1}}}{\frac{n!}{e^n}} = \lim_{n\to\infty}\frac{(n+1)!\cdot e^n}{n!\cdot e^{n+1}} = \lim_{n\to\infty}\frac{n+1}{e} = \infty$$

$\therefore$ 발산

연습문제 9.5

1. 다음 급수의 수렴성을 조사하여라.

(1) $\sum_{n=1}^{\infty}(-1)^{n+1}\frac{1}{2n+1}$

교대급수이며, $(\because (-1)^{n+1})$

$$|a_{n+1}| \le |a_n| \Rightarrow \frac{|a_{n+1}|}{|a_n|} \le 1 \Rightarrow \frac{\frac{1}{2(n+1)+1}}{\frac{1}{2n+1}} = \frac{2n+1}{2n+3}$$

$n=1$이면, $\frac{a2}{a1} = \frac{2}{5} < 1$이므로 $\{a_n\}_{n=1}^{\infty}$는 감소수열

따라서, 교대급수 판정법에 의하여

$$\lim_{n\to\infty}\frac{1}{2n+1} = 0 \Rightarrow \text{수렴}$$

$\therefore$ 수렴

(3) $\sum_{n=1}^{\infty}(-1)^{n}\frac{2n+1}{5n+1}$

교대급수이며, $(\because (-1)^{n})$

$$\frac{|a_{n+1}|}{|a_n|} \le 1 \Rightarrow \frac{\frac{2(n+1)+1}{5(n+1)+1}}{\frac{2n+1}{5n+1}} = \frac{(2(n+1)+1)\cdot(5n+1)}{(5(n+1)+1)\cdot(2n+1)} = \frac{(2n+3)(5n+1)}{(5n+6)(2n+1)}$$

$n=1$이면, $\frac{a_2}{a_1} = \frac{30}{33} < 1$이므로 $\{a_n\}_{n=1}^{\infty}$는 감소수열

따라서, 교대급수 판정법에 의하여

$$\lim_{n\to\infty}\frac{2n+1}{5n+1} = \frac{2}{5} \ne 0 \Rightarrow \text{발산}$$

$\therefore$ 발산

연습문제 9.5

3. 다음 급수의 절대수렴, 조건수렴 그리고 발산 등을 조사하여라.

(1) $\sum_{n=1}^{\infty}(-1)^{n+1}\frac{1}{3n+4}$

① 교대급수이며, $(\because (-1)^{n+1})$

$$\frac{|a_{n+1}|}{|a_n|}=\frac{\frac{1}{3(n+1)+4}}{\frac{1}{3n+4}}=\frac{3n+4}{3(n+1)+4}$$

$n=1$이면, $\frac{a_2}{a_1}=\frac{7}{10}<1$이므로 $\{a_n\}_{n=1}^{\infty}$는 감소수열

따라서, 교대급수판정법에 의하여

$\lim_{n\to\infty}\frac{1}{3n+4}=0 \Rightarrow$ 수렴

연습문제 9.6

1. 다음 함수의 n차 테일러 다항식을 구하여라.

(1) $f(x)=\cos x$, $n=5$

$$P_5(x)=f(0)+f'(0)x+\frac{f''(0)}{2!}x^2+\frac{f^{(3)}(0)}{3!}x^3+\frac{f^{(4)}(0)}{4!}x^4+\frac{f^{(5)}(0)}{5!}x^5$$

$f(x)=\cos x$ $\quad f(0)=\cos(0)=1$

$f'(x)=-\sin x$ $\quad f'(0)=-\sin(0)=0$

$f''(x)=-\cos x$ $\quad f''(0)=-\cos(0)=-1$

$f^{(3)}(x)=\sin x$ $\quad f^{(3)}(0)=\sin(0)=0$

$f^{(4)}(x)=\cos x$ $\quad f^{(4)}(0)=\cos(0)=1$

$f^{(5)}(x)=-\sin x$ $\quad f^{(5)}(0)=-\sin(0)=0$

$$\therefore P_5(x)=1+\frac{-1}{2!}x^2+\frac{1}{4!}x^4=1-\frac{1}{2}x^2+\frac{1}{24}x^4$$

(3) $f(x)=\dfrac{1}{1+x}$, $n=4$

$$P_4(x)=f(0)+f'(0)x+\frac{f''(0)}{2!}x^2+\frac{f^{(3)}(0)}{3!}x^3+\frac{f^{(4)}(0)}{4!}x^4$$

$f(x)=\dfrac{1}{1+x}=(1+x)^{-1}$ $\quad f(0)=(1+0)^{-1}=1$

$f'(x)=-(1+x)^{-2}$ $\quad f'(0)=-(1+0)^{-2}=-1$

$f''(x)=2(1+x)^{-3}$ $\quad f''(0)=2(1+0)^{-3}=2$

$f^{(3)}(x)=-3\cdot 2(1+x)^{-4}$ $\quad f^{(3)}(0)=-6(1+x)^{-4}=-6$

$f^{(4)}(x)=(-6)\cdot(-4)(1+x)^{-5}$ $\quad f^{(4)}(0)=24(1+0)^{-5}=24$

$$\therefore P_5(x)=1+(-1)x+\frac{2}{2!}x^2+\frac{(-6)}{3!}x^3+\frac{24}{4!}x^4=1-x+x^2-x^3+x^4$$

연습문제 9.7

1. 주어진 급수의 수렴구간을 구하여라.

(1) $\sum_{n=1}^{\infty} \frac{x^n}{n^2}$

수렴반경

$$R = \lim_{n\to\infty}\left|\frac{c_n}{c_{n+1}}\right| = \lim_{n\to\infty}\left|\frac{\frac{1}{n^2}}{\frac{1}{(n+1)^2}}\right| = \lim_{n\to\infty}\left|\frac{(n+1)^2}{n^2}\right|$$

$$= \lim_{n\to\infty}\frac{n^2+2n+1}{n^2} = 1$$

$x=-1 \Rightarrow \sum_{n\to1}^{\infty}\frac{(-1)^n}{n^2} = \sum_{n\to1}^{\infty} -\frac{1}{n^2}$ P급수 판정법에의하여 $2>1$ $\therefore$ 수렴

$x=1 \Rightarrow \sum_{n\to1}^{\infty}\frac{1^n}{n^2} = \sum_{n\to1}^{\infty}\frac{1}{n^2}$ P급수 판정법에의하여 $2>1$ $\therefore$ 수렴

$\therefore$ 수렴구간 : $[-1,1]$ 즉, $-1 \le x \le 1$

(4) $\sum_{n=0}^{\infty} \frac{n}{4^n} x^n$

수렴반경

$$R = \lim_{n\to\infty}\left|\frac{c_n}{c_{n+1}}\right| = \lim_{n\to\infty}\left|\frac{\frac{n}{4^n}}{\frac{n+1}{4^{n+1}}}\right| = \lim_{n\to\infty}\left|\frac{4^{n+1}(n)}{4^n(n+1)}\right|$$

$$= \lim_{n\to\infty}\frac{4n}{n+1} = 4$$

$x=-4 \Rightarrow \sum_{n\to0}^{\infty}\frac{n}{4^n}(-4)^n = \sum_{n\to1}^{\infty}(-n)$ $\therefore$ 발산

$x=4 \Rightarrow \sum_{n\to0}^{\infty}\frac{n}{4^n}(4)^n = \sum_{n\to1}^{\infty} n$ $\therefore$ 발산

$\therefore$ 수렴구간 : $(-4,4)$ 즉, $-4 < x < 4$

연습문제 9.8

1. 다음 함수의 $x=0$에서의 테일러 급수를 구하여라.

(1) $f(x)=\sin 2x$

$$f(x)=\sin 2x \qquad f(0)=0$$

$$f'(x)=2\cos 2x \qquad f'(0)=(2)(1)=2$$

$$f''(x)=-(2)2\sin 2x=-2^2\sin 2x \qquad f''(0)=-2^2(0)=0$$

$$f^{(3)}(x)=-(2)2^2\cos 2x=-2^3\cos 2x \qquad f^{(3)}(0)=-2^3(1)=-2^3$$

$$f^{(4)}(x)=(2)2^3\sin 2x=2^4\sin 2x \qquad f^{(4)}(0)=-2^4(0)=0$$

$$f^{(5)}(x)=(2)2^4\cos 2x=2^5\cos 2x \qquad f^{(5)}(0)=2^5(1)=2^5$$

$$\vdots \qquad\qquad \vdots$$

$$f(x)=f(0)+f'(0)x+\frac{f''(0)}{2!}x^2+\frac{f^{(3)}(0)}{3!}x^3+\cdots+\frac{f^{(n)}(0)}{n!}x^n$$

$$=\cancel{0}+2x+\cancel{\frac{2^2(0)}{2!}}x^2-\frac{2^3(1)}{3!}x^3+\cancel{\frac{2^4(0)}{4!}}x^4+\frac{2^5(1)}{5!}x^5+\cdots$$

$$=\sum_{k=0}^{\infty}\frac{(-1)^k 2^{2k+1}}{(2k+1)!}x^{2k+1}$$

$$\therefore \sum_{k=0}^{\infty}\frac{(-1)^k(2x)^{2k+1}}{(2k+1)!} \qquad (\text{예제}9.39(ii)\text{참고})$$

연습문제 9.8

3. 다음 함수의 $x=a$에서의 테일러 급수를 구하여라.

(1) $f(x)=x^4-x+2, \quad a=-1$

a에 대한 f의 n차 테일러 다항식

$$P_n(x)=f(a)+f'(a)(x-a)+\frac{f''(a)}{2!}(x-a)^2+\cdots+\frac{f^{(n)}(a)}{n!}(x-a)^n$$

$f(x)=x^4-x+2 \qquad f(-1)=4$

$f'(x)=4x^3-1 \qquad f'(-1)=-5$

$f''(x)=12x^2 \qquad f''(-1)=12$

$f^{(3)}(x)=24x \qquad f^{(3)}(-1)=-24$

$f^{(4)}(x)=24 \qquad f^{(3)}(-1)=24$

$n \geq 5$이면 $f^{(n)}(-1)=0$

$$\therefore\ f(x)=4-5(x+1)+\frac{12}{2!}(x+1)^2-\frac{24}{3!}(x+1)^3+\frac{24}{4!}(x+1)^4$$

$$=4-5(x+1)+6(x+1)^2-4(x+1)^3+(x+1)^4$$